



Green University of Bangladesh
Department of Computer Science and Engineering (CSE)
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Mini Bank Management System

Course Title: Object Oriented Programming Lab
Course Code: CSE 202 Section: 251 D1

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Chapter 1

Introduction

1.1 Overview

The Mini Bank Management System is a simple software project developed using Java and Object Oriented Programming concepts. The main purpose of this system is to simulate basic banking operations such as creating an account, depositing money, withdrawing money, checking balance, and viewing transaction history.

This project also includes a graphical user interface (GUI) to make the system user-friendly. The system is designed in a simple way so that users can easily understand and use it. It shows how real banking systems work in a simplified form.

1.2 Motivation

The main reason for choosing this project is to understand how banking systems work in real life. Banking is an important part of our daily life, and almost everyone uses banking services.

This project helps to learn important programming concepts such as classes, objects, inheritance, and encapsulation. It also helps to understand how to design a real-life system using programming. Another motivation is to improve problem-solving skills and GUI design skills.

1.3 Problem Definition

1.3.1 Problem Statement

In real life, managing bank accounts manually is difficult and time-consuming. There is a need for a system that can store customer information, manage transactions, and provide quick services.

The problem is to design a system that can:

- Create and manage user accounts
- Handle deposits and withdrawals
- Maintain transaction history
- Provide secure login using PIN

This system should be simple, efficient, and easy to use.

1.3.2 Complex Engineering Problem

Table 1.1: Summary of the attributes touched by the mentioned projects

Name of the P Attributes	Explain how to address
P1: Depth of knowledge required	Requires knowledge of Java, OOP, and GUI design.
P2: Range of conflicting requirements	Needs both security and simplicity for users.
P3: Depth of analysis required	Must analyze user actions like deposit, withdraw, and validation.
P4: Familiarity of issues	Banking system concepts are common but implementation is challenging.
P5: Extent of applicable codes	Uses multiple classes and methods for different operations.
P6: Extent of stakeholder involvement and conflicting requirements	Users want easy interface, system needs secure handling.
P7: Interdependence	Account, transaction, and user data depend on each other.

1.4 Design Goals/Objectives

The main objectives of this project are:

- To create a simple banking system using Java
- To implement Object Oriented Programming concepts
- To design a user-friendly graphical interface
- To provide secure login using PIN
- To manage account balance and transactions correctly
- To store and display transaction history

The system should be reliable, easy to use, and efficient.

1.5 Application

The Mini Bank Management System can be used in many real-life situations. It can be used in small banks, educational purposes, and training systems. This project helps students understand how banking software works.

It can also be used as a base system for developing more advanced banking applications. For example, features like online banking, ATM systems, and mobile banking apps can be added in the future.

In real life, similar systems are used in banks to manage millions of customer accounts. This project represents a simplified version of those systems and helps to understand their basic working process